

Prediction of Silicon Content of Hot Metal in Blast Furnace Based on Optuna-GBDT

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The silicon content of hot metal is a key index for the determination of blast furnace status, and accurate prediction of the silicon content of hot metal is crucial for blast furnace ironmaking. First, 10992 sets of blast furnace data obtained from the site of an iron and steel enterprise were preprocessed. Then, 22 important feature parameters related to the silicon content of hot metal were screened by feature engineering. Finally, the hyperparameters of the Gradient Boosting Decision Tree (GBDT) algorithm model were optimized with the help of the Optuna framework, and the Optuna-GBDT model was established to predict the silicon content of hot metal. The experimental results show that compared with the Bayesian algorithm and the traditional stochastic search method, the Optuna framework can achieve better hyperparameter optimization with fewer iterations and smaller errors. The Optuna-GBDT model performs better in predicting the silicon content of hot metal compared with the optimized Random Forest (RF), Decision Tree and AdaBoost models, and the prediction results are basically in line with the actual values, with the mean absolute error (MAE) of 0.0094, the root mean square error (RMSE) of 0.0152, and the coefficient of determination (R²) of 0.975. The experimental results verified the validity and feasibility of establishing the Optuna-GBDT model to predict the silicon content of hot metal, which provides a reliable tool for iron and steel enterprises and helps to optimize the ironmaking process, improve production efficiency and product quality.

KEY WORDS: silicon content of hot metal; GBDT model; Optuna; data processing.

1. Introduction

With the development of the modern iron and steel industry, the blast furnace, as one of the most important pieces of ironmaking equipment, plays a key role in ensuring continuous production and product quality. Accurate prediction of the silicon content of hot metal is crucial for the determination of blast furnace condition, optimization of the smelting process, and assurance of final product quality.¹⁾ In recent years, with the rapid development of machine learning and artificial intelligence technologies, the application of intelligent models for the prediction of the silicon content of hot metals has become a hot area of research. By processing and analyzing a large amount of real-time data, intelligent models are able to learn the complex associations between the data and generate accurate prediction results.

In a related study, Gradually Ling *et al.* used the support vector machine model to realize the silicon content of hot metal, multi-class classification, and evaluated the perfor-

mance of each classifier.²⁾ Shi *et al.* developed a predictive regression model for the silicon content of hot metals using partial least squares regression.³⁾ Yu *et al.* predicted the silicon content by establishing a classification regression tree and a support vector machine prediction model and found that the classification regression tree model has higher prediction accuracy.⁴⁾ Zhao *et al.* found that the random forest model has a better prediction of blast furnace gas utilization by comparing three intelligent models, such as the Adaboost model, the random forest model, and the neural network model.⁵⁾ Liu *et al.* compared the performance of the Random Forest Model, Decision Tree Model, and Ada-boost Model in the prediction of the silicon content of hot metal, and the results showed that the prediction accuracy of the Adaboost Model was higher.⁶⁾

In addition, some optimized and combined prediction models have been proposed. The LSTM-RNN model was used by Li *et al.* to predict the silicon content of hot metal, and the results showed that it had a more accurate prediction than the RNN model.⁷⁾ Jian *et al.* accomplished the prediction of silicon content of hot metal in a blast furnace

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using a least squares support vector machine (LS-SVM) with a certain degree of accuracy.⁸⁾ Yang proposed a new model combining empirical modal decomposition (EMD), genetic algorithm (GA), and long short-term memory neural network (LSTM), which was confirmed to be superior to the support vector regression (SVR) model and the standard LSTM neural network model.⁹⁾ Cui *et al.* used the GWO-SVR model to predict each state variable sequentially and applied the prediction results to the dynamic prediction of the silicon content of hot metal.¹⁰⁾ Diniz, A.P.M. *et al.* combined wavelet decomposition with a neural network model for the prediction of the silicon content of hot metal and controlled the prediction error within an acceptable range.¹¹⁾ H.G. Zhang *et al.* established a neural network model for predicting the silicon content in iron water through the Extreme Learning Machine (ELM) algorithm and used the Weighted ELM (W-ELM) algorithm to predict the trend of silicon content.¹²⁾ Li *et al.* determined the feature variables of the prediction model by combining the feature selection with the theory of ironmaking and verified the feature engineering and stacking algorithms for the blast furnace parameter prediction.¹³⁾ Fang *et al.* proposed a multicore extreme learning machine (MKELM) optimized by the Improved Grey Wolf Algorithm (IGWO) for predictive modeling of the silicon content of hot metal in blast furnaces and found experimentally that it outperformed the backpropagation neural network (BP-NN), extreme learning machine (ELM), KELM, and GWO-MKELM.¹⁴⁾

Given the problems encountered in the above re-search: (1) The small amount of feature parameter data does not satisfy the problem of model training. (2) The unreasonable selection of characteristic parameters affects the accurate prediction of the model. (3) the problem of improper selection of the prediction model parameters, which makes the model ineffective. For this reason, 10992 sets of data from the blast furnace ironmaking site of an iron and steel enterprise are used for data preprocessing, feature selection by Pearson correlation analysis and mutual information method, and optimization of model hyperparameters by the Optuna framework to establish a prediction model for the silicon content of hot metal.

2. Blast Furnace Data Preprocessing

2.1. Gradient Boosting Decision Tree Principle

GBDT (Gradient Boosting Decision Tree) is a decision tree-based gradient boosting algorithm that continuously improves the predictive ability of the model by iteratively training the decision tree.¹⁵⁾ In this, each decision tree is constructed to reduce the residuals of the previous model and make the final residuals converge to 0 in the gradient direction by iterating continuously. The schematic diagram of GBDT is shown in Fig. 1.

The GBDT regression algorithm is implemented as follows:

Input training set $T = \{(x_1, y_1), (x_2, y_2), (x_i, y_i), \dots, (x_N, y_N)\}$, where $x_i \in X \subseteq R^n$, X is the input sample space. x_i is the input features of hot metal. $y_i \in Y \subseteq R^n$, Y is the silicon content of hot metal.

(1) Initializing the first learner $f_0(x)$:

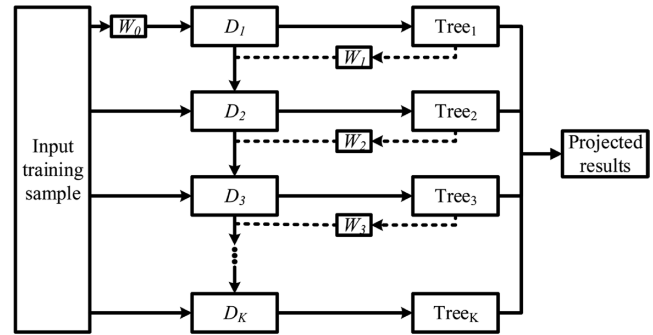


Fig. 1. GBDT prediction principle.

$$f_0(x) = \operatorname{argmin}_i \sum_i^N L(y_i, c) \dots \dots \dots (1)$$

In the equation: $L(y_i, c)$ for the loss function. c a constant to minimize the loss function.

(2) Establishing the M classification regression tree: $m = 1, 2, \dots, M$:

For samples $i = 1, 2, \dots, N$, compute the residual corresponding to the m -th tree:

$$r_{m,i} = - \left[\frac{\partial L(y_i, f(x_i))}{\partial f(x)} \right]_{f(x)=f_{m-1}(x)} \dots \dots \dots (2)$$

Using the regression tree to fit the data $(x_i, r_{m,i})$, the leaf node region $R_{m,j}$ corresponding to the m -th tree is obtained, where $j = 1, 2, \dots, J_m$.

For $j = 1, 2, \dots, J_m$, calculating the best fit:

$$c_{m,j} = \operatorname{argmin}_{x_i \in R_{m,j}} \sum L(y_i, f_{m-1}(x_i) + c) \dots \dots \dots (3)$$

Updating for the Strong Learner $f_m(x)$:

$$f_m(x) = f_{m-1}(x) + \sum_{j=1}^{J_m} c_{m,j} I(x \in R_{m,j}) \dots \dots \dots (4)$$

The expression for the final strong learner $f_M(x)$ is obtained:

$$f_M(x) = f_0(x) + \sum_{m=1}^M \sum_{j=1}^{J_m} c_{m,j} I(x \in R_{m,j}) \dots \dots \dots (5)$$

2.2. Optuna Framework Optimization Parameters

Optuna is a software framework specifically designed for machine learning to automatically search for the best hyperparameter configuration. A technique called Sequential Model-based Optimization (SMBO) was used to optimize the process.¹⁶⁾ By analyzing the tried hyperparameter configurations and their corresponding performance metrics, Optuna constructs a model to predict better hyperparameter configurations. Then, it selects the next hyperparameter configuration to try based on the model's prediction. By iterating this process duplicatedly, Optuna gradually searches for the hyperparameter configuration with the best performance. The optimization process of the Optuna framework is shown in Fig. 2.

The specific steps are as follows:

Step 1: Defining spatial parameters, the range and type of hyperparameters, such as continuous, discrete, or categori-

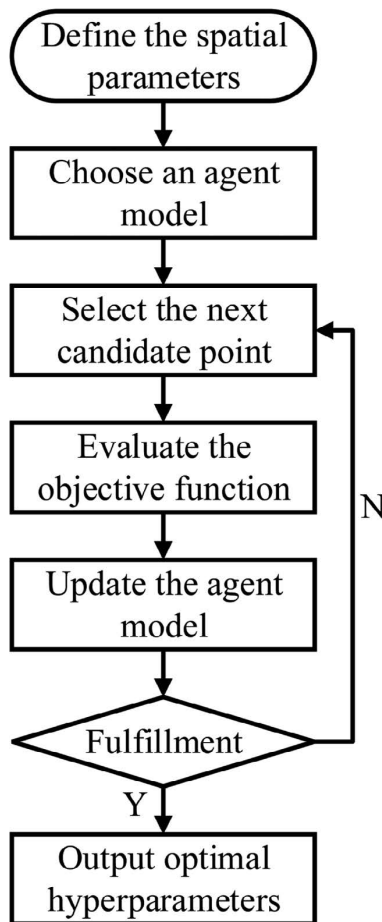


Fig. 2. Optuna optimization process.

cal parameters, and setting the range of values are required before optimizing the model.

Step 2: Optuna uses a Gaussian process as a proxy model for estimating the performance of the objective function and calculating the uncertainty.

Step 3: Selecting the next candidate at each iteration, a strategy called “sampler” is used to select the next combination of hyperparameters to be evaluated based on the predictions and uncertainties of the agent model.

Step 4: Evaluating the objective function using the selected combination of hyperparameters on the objective function and calculate an evaluation value.

Step 5: Updating the agent model by using the new combination of hyperparameters and the corresponding evaluation values to update the agent model so that it can more accurately predict the performance of the objective function.

Step 6: Iterative process, duplicate steps 3–5 until the specified number of iterations is reached or the termination condition is satisfied.

Step 7: Based on the optimized agent model, the combination of hyperparameters found with the best performance is output as the final result.

2.3. Forecasting Process

The process of predicting the silicon content of hot metals using the Optuna-GBDT model is shown in Fig. 3.

The specific steps are as follows:

Step 1: Obtaining real data on the production of iron from an iron and steel company.

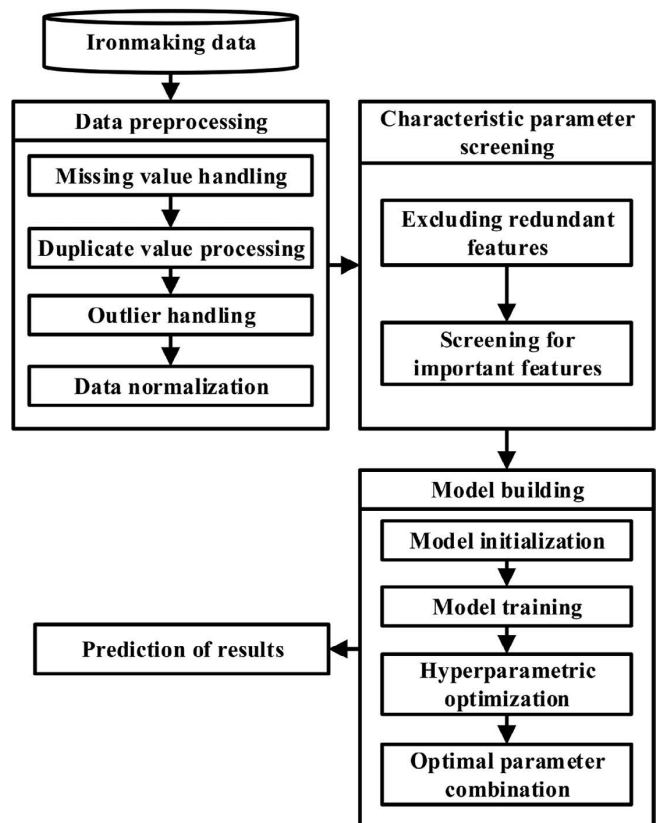


Fig. 3. Silicon content of hot metal prediction process.

Step 2: Preprocessing of the collected data, including the handling of missing values, duplicate values, outliers, and the normalization of the data.

Step 3: The preprocessed data is imported into the feature parameter screening framework. Redundant features are first removed, and then the remaining features are screened for importance.

Step 4: Using the filtered feature parameters, construct a gradient-boosting decision tree (GBDT) prediction model and use the Optuna framework to optimize the hyperparameters of the GBDT to find the optimal parameter combination.

Step 5: The GBDT model constructed using the optimal parameter combination is used for prediction, and the final output is obtained.

3. Data Collection

The data were selected from the production data and calculation data of a steel enterprise from May 2019 to August 2020, totaling 10992 sets of parameter data. The sampling frequency of the collected real-time data is 1 h, and the frequency of the inspection and assay data is batch. In order to ensure the unity of the data frequency, the inspection and assay data parameters need to be adjusted. Through the amplification of the inspection and assay data, using Python to carry out the time integer processing, we obtained 1 h as the frequency of 48 data parameters (including the target parameters), as shown in Table 1. The results of the slag iron composition of the inspection test were integrated with the real-time data 2 h in advance to achieve the goal of predicting the silicon content of hot metal 2 h in advance.

Table 1. Parameter data.

Operating parameter		Inspecting parameter
Wind temperature (WT)	Hydrogen (H ₂)	Iron temperature (IT)
Wind pressure (WP)	Nitrogen (N ₂)	Si
Total oxygen (TO)	Air volume (AV)	Mn
Oxygen enrichment ratio (OER)	Inlet air volume (IAV)	P
Top temperature (TT)	Pulverized coal (PC)	C
Top pressure (TP)	Water flow rate (WFR)	S
Breathability index (BI)	Incoming water temperature (IWT)	Ti
Differential pressure (DP)	Return water temperature (RWT)	SiO ₂
Standard wind speed (SWS)	Furnace bottom flow (FBF)	CaO
Actual wind speed (AWS)	Furnace bottom temperature difference (FBTD)	MgO
kinetic energy (KE)	Seal chamber N ₂ (SCN ₂)	Al ₂ O ₃
Edge Temperature (ET)	Seal chamber temperature (SCT)	MnO
carbon dioxide (CO ₂)	Seal chamber water volume (SCWV)	FeO
Carbon Monoxide (CO)	Furnace belly gas (FBG)	
CH ₄	Cold air temperature (CAT)	
Lower Differential Pressure (LDP)	Net gas content (NGC)	
Upper Differential Pressure (UDP)	Barren gas content (BGC)	
TRT		

4. Data Processing

During the blast furnace smelting process, abnormal or missing production data may occur due to abnormal furnace conditions or instrument failures.¹⁷⁾ In order to ensure the quality and accuracy of the data to meet the prediction needs, the acquired data are processed before loading the data, including missing values, duplicate values, outliers, and normalization of the data.¹⁸⁾

4.1. Missing Value Handling

During the production process, data collection failures and missing values may occur due to machine failures or human factors. **Figure 4** is a visualization of the missing values for the five parameters listed, which shows the number of missing values and their percentage in the raw data. In order to deal with these missing values, different approaches are used for different types of data. Facing discrete data (*e.g.*, test parameters) is filled using proximity values, and facing continuous data (*e.g.*, operational parameters) is filled using Lagrangian interpolation.¹⁹⁾

4.2. Duplicate Value Processing

Duplicate values refer to the presence of identical or nearly identical records in a data set. Dealing with duplicate values is an important part of data processing to avoid introducing biases and errors in the data analysis and modeling processes. When records with identical parameters are present in a dataset, these duplicates may be due to errors or duplicated operations during data collection, transmission, or entry. By removing duplicate data, the uniqueness of each record can be ensured, avoiding erroneous conclusions during statistical analysis and modeling. In addition, deleting duplicate data can improve the efficiency and readability

of data sets and reduce unnecessary redundant information.

4.3. Outliers Handling

Outliers are observations in the data set that are significantly different from the majority of the sample. Due to the harsh production environment of the blast furnace, the data may have outliers.²⁰⁾ In order to ensure the accuracy of the prediction results, the outliers are handled using a box-and-line plot.²¹⁾ A box-and-line plot is used to understand the concentration trend, degree of dispersion, and presence of outliers in the data by displaying statistics such as the median, upper and lower quartiles, and maximum and minimum values of the data. Below is the definition of box-and-line plot boundaries:

$$UB = Q_3 + 1.5IQR \quad \text{..... (6)}$$

$$LB = Q_1 - 1.5IQR \quad \text{..... (7)}$$

$$IQR = Q_3 - Q_1 \quad \text{..... (8)}$$

The above equation: *UB* represents the upper boundary, *LB* represents the lower boundary, *Q₁* and *Q₃* represent the upper quartile and lower quartile, respectively, and *IQR* is the interquartile range, *i.e.*, the difference between the upper and lower quartiles. Between the upper and lower boundaries is the normal value area, and the data outside the boundaries are the outliers that need to be deleted.

Taking the feature parameter [Ti] as an example, the boxplots are processed, and the boxplots before and after processing are shown in **Fig. 5**. The boxplot before processing shows the original distribution of the data, where data outside the upper and lower boundaries are recognized as outliers. The upper and lower boundaries are calculated according to Eqs. (6), (7), and (8), and the data points

beyond the boundaries are labeled as outliers. In the processed boxplot, it can be observed that the outliers have been removed and the boxes of data are more centralized, with no obvious outliers present.

4.4. Data Normalization

Data normalization, also known as standardization, is the process of converting data with different ranges and units into a uniform standard range. Data normalization can eliminate the differences in scale between different features, making the data more comparable and interpretable.²²⁾ Before predicting the parameters, the data need to be normalized to avoid the problem of inconsistency in the parameter units of the data, resulting in large order of magnitude differences between features. The Z-score normalization method is used to normalize the data, which has the characteristics of simple and fast arithmetic and strong anti-interference. The

equation is as follows:

$$x' = \frac{x - \mu}{\sigma} \dots\dots\dots (9)$$

where x' denotes standardized value, x denotes original value, μ denotes overall mean value, σ denotes overall standard deviation.

5. Characteristic Parameter Screening

Before model building, feature parameter screening is performed to select the most suitable input parameters to improve the efficiency of the model, reduce the computation time, and enhance the predictive ability of the model. Blast furnace smelting is a complex process in which there are close correlations and uncertainties among different parameters. Therefore, it is extremely important to select appropriate inputs for characteristic parameters when building models for prediction.

5.1. Pearson Correlation Analysis

The Pearson correlation coefficient is a commonly used linear correlation coefficient to measure the degree of correlation between two variables. The value ranges from -1 to 1 , greater than 0 means positive correlation, less than 0 means negative correlation, and the larger the absolute value, the stronger the correlation.^{23,24)} Its calculation equation is as follows:

$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} \dots\dots\dots (10)$$

where $\rho_{X,Y}$ is the overall Pearson correlation coefficient of variables X and Y , $\text{cov}(X,Y)$ is the covariance between variables X and Y , σ_X is the standard deviation of variable X ; and σ_Y is the standard deviation of variable Y .

The different parameters in the processed dataset were selected sequentially and denoted as X_i and Y_i ($i = 1, 2, \dots, n$), and the covariance and standard deviation estimation were performed to obtain specific Pearson's correlation

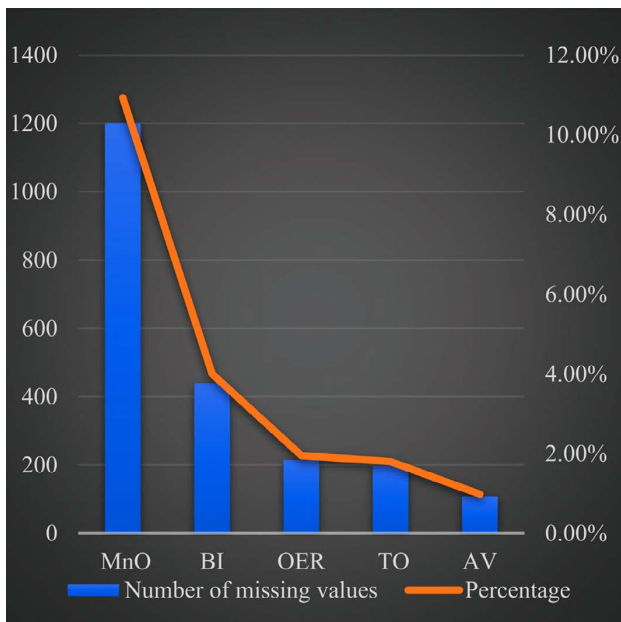


Fig. 4. Missing value visualization. (Online version in color.)

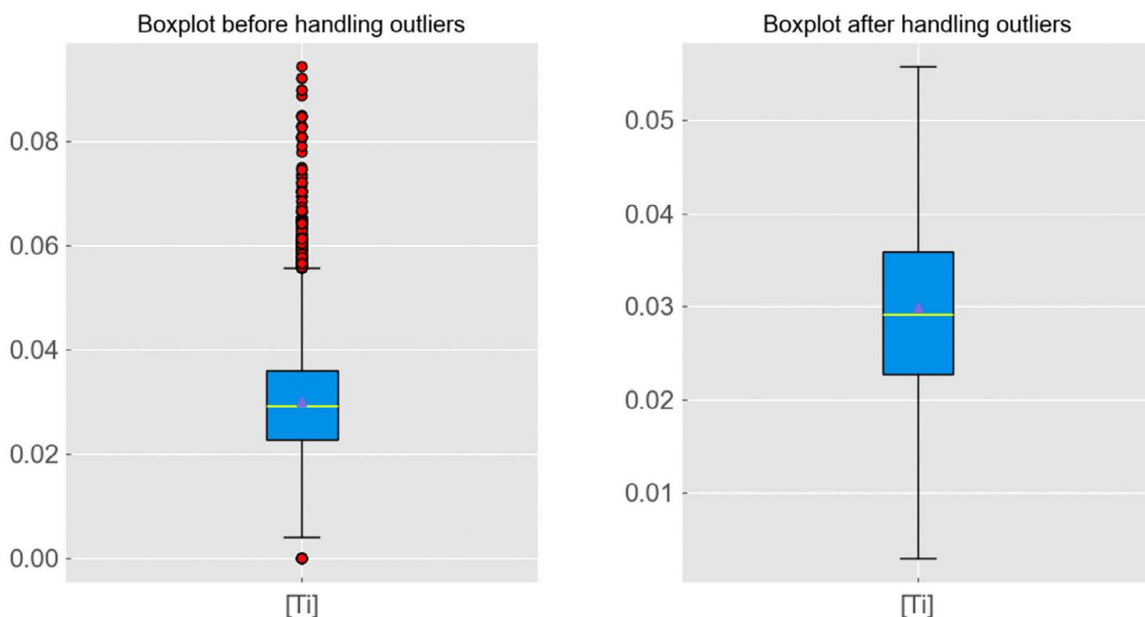


Fig. 5. About [Ti] Boxplots. (Online version in color.)

coefficients with the following expressions:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \dots\dots\dots (11)$$

The correlation between the 48 parameters was analyzed by plotting a heat map of the Pearson correlation coefficient matrix. See **Fig. 6**. The correlation between the parameters was assessed based on the color, with bluer colors indicating a higher correlation and redder colors indicating a lower correlation. It is observed that a total of 8 redundant parameters such as (SiO₂), (CaO), (MgO), standard wind speed, wind volume, air inlet, wind pressure, and kinetic energy should be deleted to reduce the burden of the machine learning algorithm and simplify the algorithm model to improve the efficiency and accuracy of the algorithm.

5.2. Mutual Information Method Eigenvalue Screening

Mutual Information (MI) is a concept used in information theory to measure the dependence or correlation between two random variables, measuring the ability of the information in one variable to explain the other variable.²⁵⁾ The definition of mutual information is based on the concept of information entropy. For two discrete random variables X and Y , mutual information (MI) is defined as follows:

$$MI(X, Y) = H(X) + H(Y) - H(X, Y) \dots\dots\dots (12)$$

where $H(X)$ and $H(Y)$ denote the information entropy of random variables X and Y , respectively, and $H(X, Y)$ denotes the information entropy of two random variables X and Y jointly.

The information entropy reflects the uncertainty or unpredictability of the random variable X . When the information entropy is large, it indicates that the value of the random variable X has high uncertainty; when the information entropy is small, it indicates that the value of the random variable X has low uncertainty, *i.e.*, it is easier to predict.²⁶⁾

Information entropy is defined as follows:

$$H(X) = -\sum p(x) \log_b p(x) \dots\dots\dots (13)$$

where $p(x)$ denotes the probability that the random variable X takes a certain value x . For a discrete random variable X , the probability distribution is computed by multiplying the corresponding probability $p(x)$ for each value x taken, then taking the logarithm and summing, and finally taking the negative value to obtain the information entropy.

The larger the value of mutual information, the stronger the correlation between the two variables; the smaller the value or close to zero, the weaker or irrelevant the correlation between the two variables. From **Fig. 7**, it can be observed that the magnitude of importance of each feature for the prediction of the silicon content of hot metal is shown in order.

According to the process of blast furnace ironmaking combined with the order of importance shown in **Fig. 7**, excluding the characteristic parameters with parameter information correlation coefficients less than 0.1, the finalized characteristic parameters of the input model are [Ti], [C], [S], [MnO], [P], [Mn], (Al₂O₃), (FeO), molten iron temperature, top pressure, sealing chamber water volume,

bottom flow rate, net gas content, cold air temperature, furnace bottom temperature difference, pulverized coal, total oxygen, sealing chamber nitrogen, water flow, wind temperature, oxygen enrichment rate, upper pressure difference, *etc.*, a total of 22 characteristic parameters.

6. Model Prediction and Experimental Analysis

The establishment of the model needs to divide the data set into a training set and a test set. The training set is used to train and learn the model, and the test set is used to test the learning effect. Considering the continuity of blast furnace production, all data sets are divided in a ratio of 8:2 in chronological order, and 20% of the data is taken as the test set to test the model to evaluate its performance.

6.1. Guidelines for Optimization and Assessment

The model optimization criteria and model performance evaluation are based on the following: The coefficient of determination, R², which measures the degree of model fit to the observed data, ranges from 0 to 1. The closer R² is to 1, the better the model fits the data, and the smaller the corresponding three evaluation parameters, such as the mean squared error (MSE), the root mean squared error (RMSE), and the mean absolute error (MAE), are, the better the model fits the data.

MSE is the square of the difference between the actual and predicted values, then summed and averaged.

$$MSE = \frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2 \dots\dots\dots (14)$$

In the equation, x_i is the predicted value, y_i is the actual value, and n is the sample size.

RMSE is the ratio of the square of the deviation of the predicted value from the actual value to the overall sample, with the same parameters as in the above equation.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2} \dots\dots\dots (15)$$

MAE is the average of the absolute errors of the actual and predicted values, where the parameters of the equation are the same as above.

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_i - y_i| \dots\dots\dots (16)$$

The coefficient of determination R² takes the mean as the error benchmark to determine the relationship between the prediction error and the mean benchmark error.

$$R^2 = \frac{\sum_{i=1}^N (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^N (y_i - \bar{y})^2} = 1 - \frac{SSE}{SST} \dots\dots\dots (17)$$

In the equation: y_i is the value to be fitted, the mean value is $\bar{y}$, the fitting value is $\hat{y}_i$, *SSE* is the sum of residual squares, and *SST* is the total average.

6.2. Optuna Framework Optimization

Optuna is used to optimize the hyperparameters of the GBDT algorithm, and the parameters of the GBDT algorithm are shown in **Table 2**.

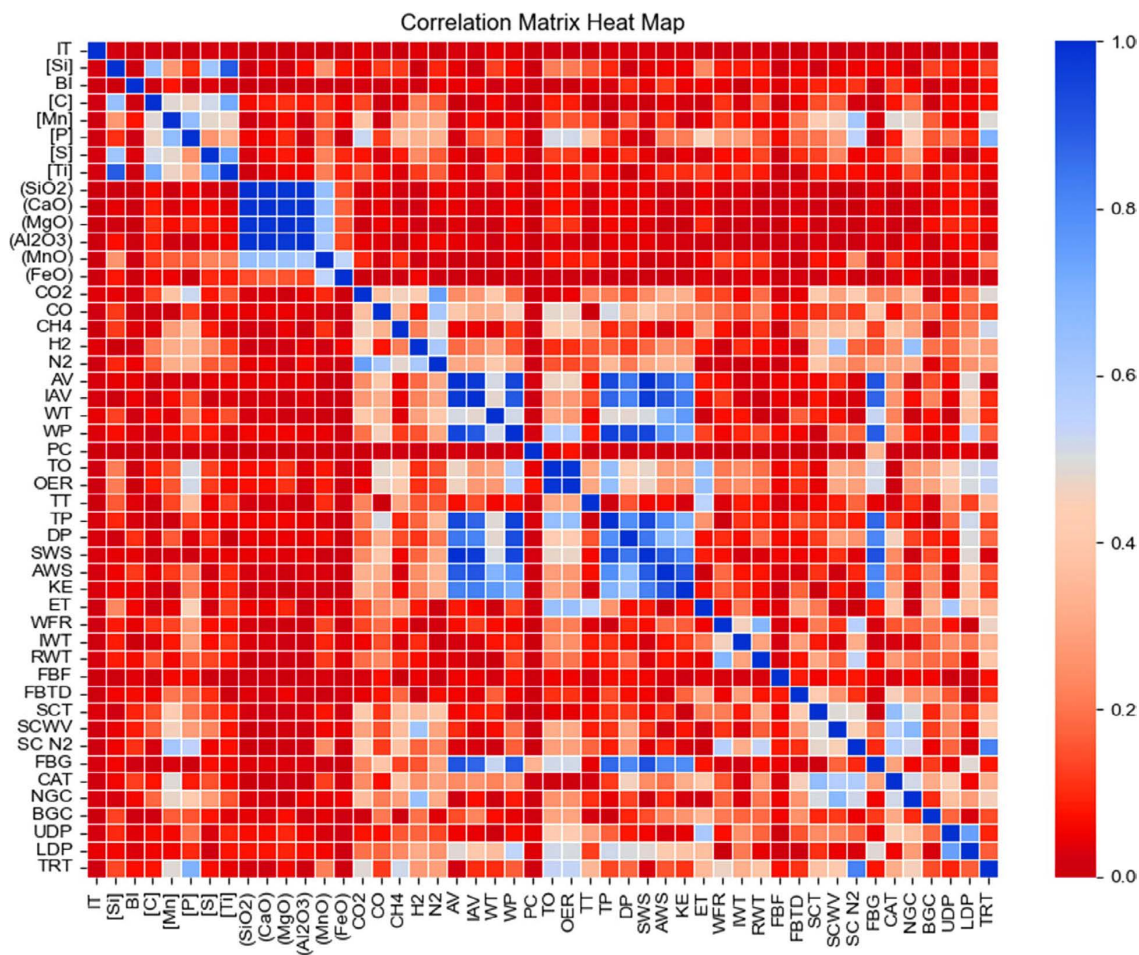


Fig. 6. Heat map of the Pearson correlation matrix. (Online version in color.)

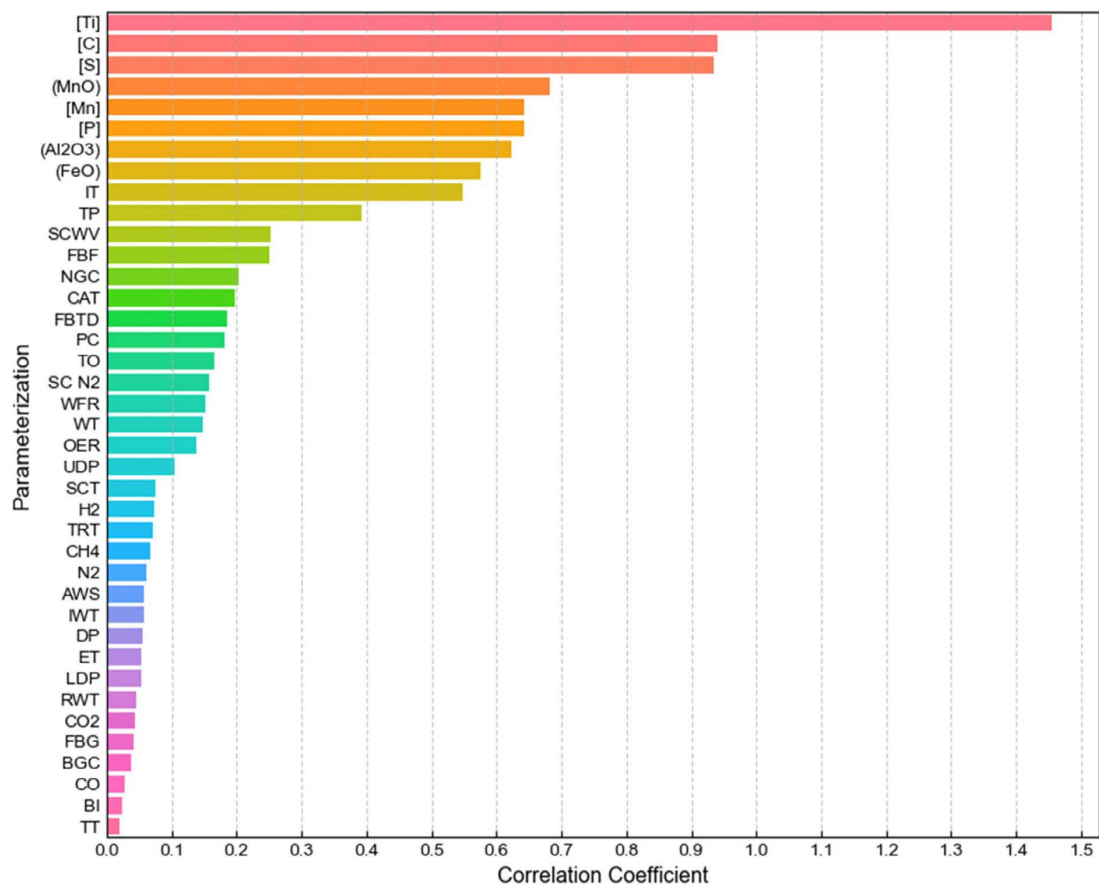


Fig. 7. Importance ranking of feature parameters. (Online version in color.)

In order to show the effect of Optuna in the optimization of parameters of the GBDT algorithm model, the parameter optimization of the GBDT algorithm model is done by using random search,²⁷⁾ Bayesian methods,²⁸⁾ and Optuna, respectively, to do comparative experiments. The results of the experiments are shown in **Table 3**, and it is observed that Optuna has fewer iterations, less error, and higher accuracy than the traditional random search method and Bayesian algorithm.

Optuna’s optimization process for the hyperparameters of the GBDT algorithm is shown in **Fig. 8**. It can be observed that the R2 score stabilizes at the 10th iteration and achieves a maximum value of 0.975 at the 19th iteration. The optimal parameter combinations corresponding to the optimization results are shown in **Table 4**.

6.3. Model Prediction Assessment

6.3.1. Comparison Algorithm Selection

The Decision Tree algorithm, which is frequently used for regression prediction, Random Forest (RF), and Adaptive Enhancement (AdaBoost) algorithms were chosen as

Table 2. GBDT algorithm parameters.

Parameters	Range of values
n_estimators (N_E)	(50, 300)
learning_rate (L_R)	(0.001, 0.2)
max_depth (M_D)	(3, 10)
subsample (SP)	(0.6, 1.0)
max_features (M_F)	(0.1, 1.0)
min_samples_split (M_S_S)	(2, 20)
min_samples_leaf (M_S_L)	(1, 10)

Table 3. GBDT optimization results comparison.

Optimization methods	Iteration number	MAE	RMSE	R2
GBDT		0.0182	0.024	0.934
RS-GBDT	254	0.0102	0.159	0.973
BM-GBDT	85	0.0102	0.016	0.973
Optuna-GBDT	19	0.0098	0.015	0.975

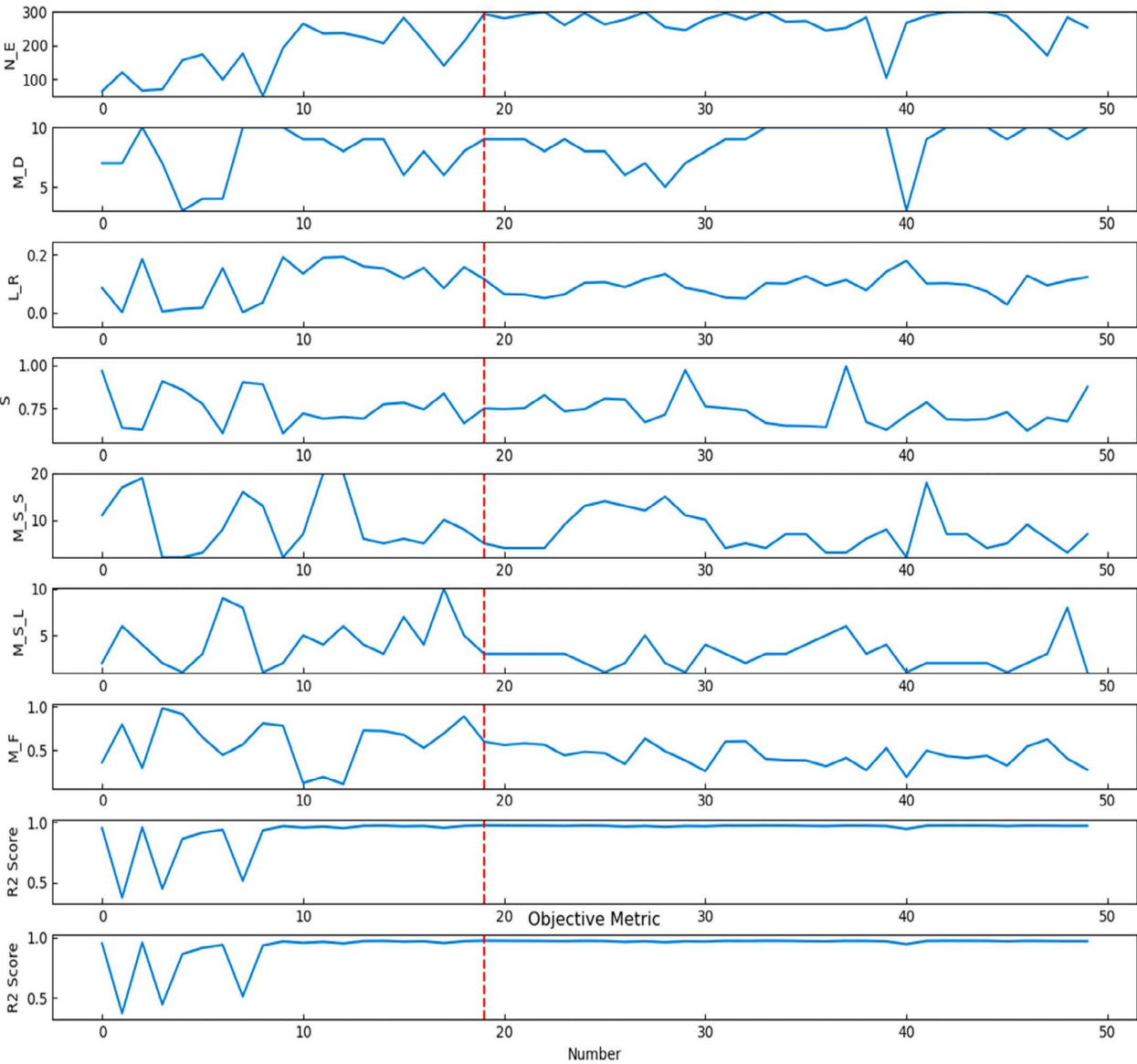


Fig. 8. Optuna parameter optimization process. (Online version in color.)

comparison algorithms to build the prediction model of the silicon content of hot metal in order to assess the efficacy of the Optuna-GBDT prediction model.

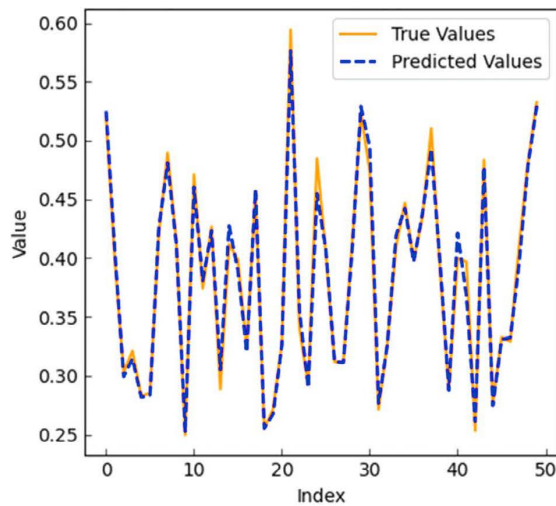
The Decision Tree algorithm is based on a branching structure based on conditional statements by recursively dividing the dataset into different subsets, which ultimately forms a tree-like structure.^{29,30)} The Decision Tree algorithm divides the dataset internal node represents a dividing fea-

ture, each branch represents a value taken from that feature, and each leaf node represents a category or a predicted value. The maximum depth of the tree is 10, the minimum number of samples required to perform a node split is 7, and the minimum number of samples required on a leaf node is 2, as confirmed by the Optuna framework.

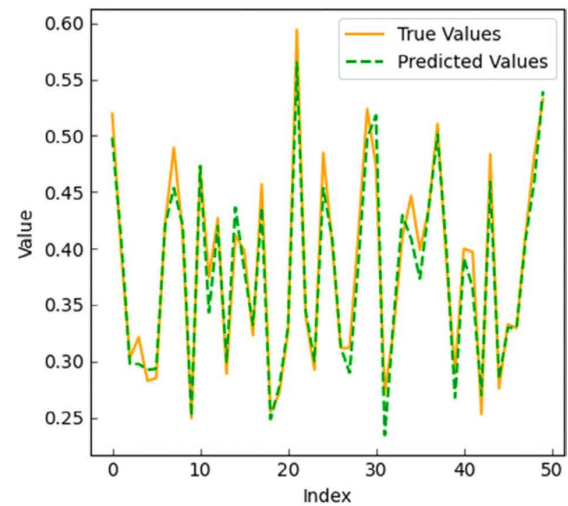
The Random Forest (RF) algorithm is an integrated learning algorithm based on decision trees. Random Forests consist of multiple decision trees, each of which independently trains and predicts the data.^{31,32)} In constructing each decision tree, Random Forest uses Bootstrap Sampling (Bootstrap Sampling) to take samples from the original dataset and put them back to generate different training sets. At the same time, in the node division process of the decision tree, Random Forest also introduces feature random selection, *i.e.*, a portion of features from all features are randomly selected to participate in node division. According to the Optuna framework, it is confirmed that the number of trees is 129, the maximum depth is 10, the minimum number of samples required for node division is 11, and the minimum number of samples required for leaf nodes is 6.

Table 4. GBDT algorithm parameters.

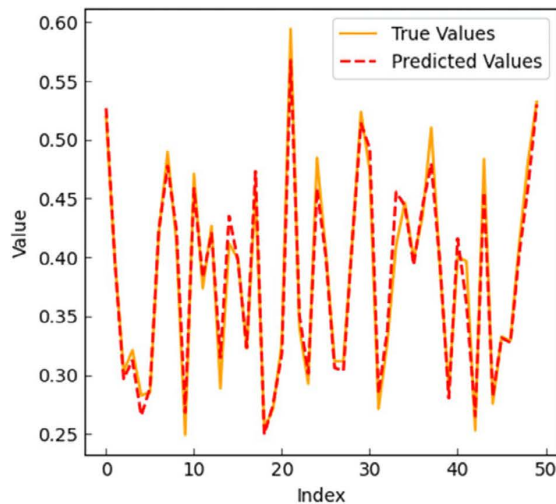
GBDT Parameters	
n_estimators (N_E)	293
learning_rate (L_R)	0.12
max_depth (M_D)	9
subsample (SP)	0.75
max_features (M_F)	0.59
min_samples_split (M_S_S)	5
min_samples_leaf (M_S_L)	3



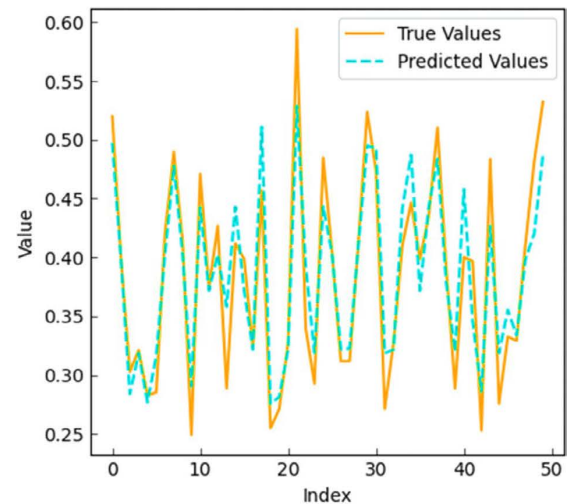
(a) GBDT



(b) DecisionTree



(c) Random Forest



(d) AdaBoost

Fig. 9. Actual and predicted silicon content of hot metal. (Online version in color.)

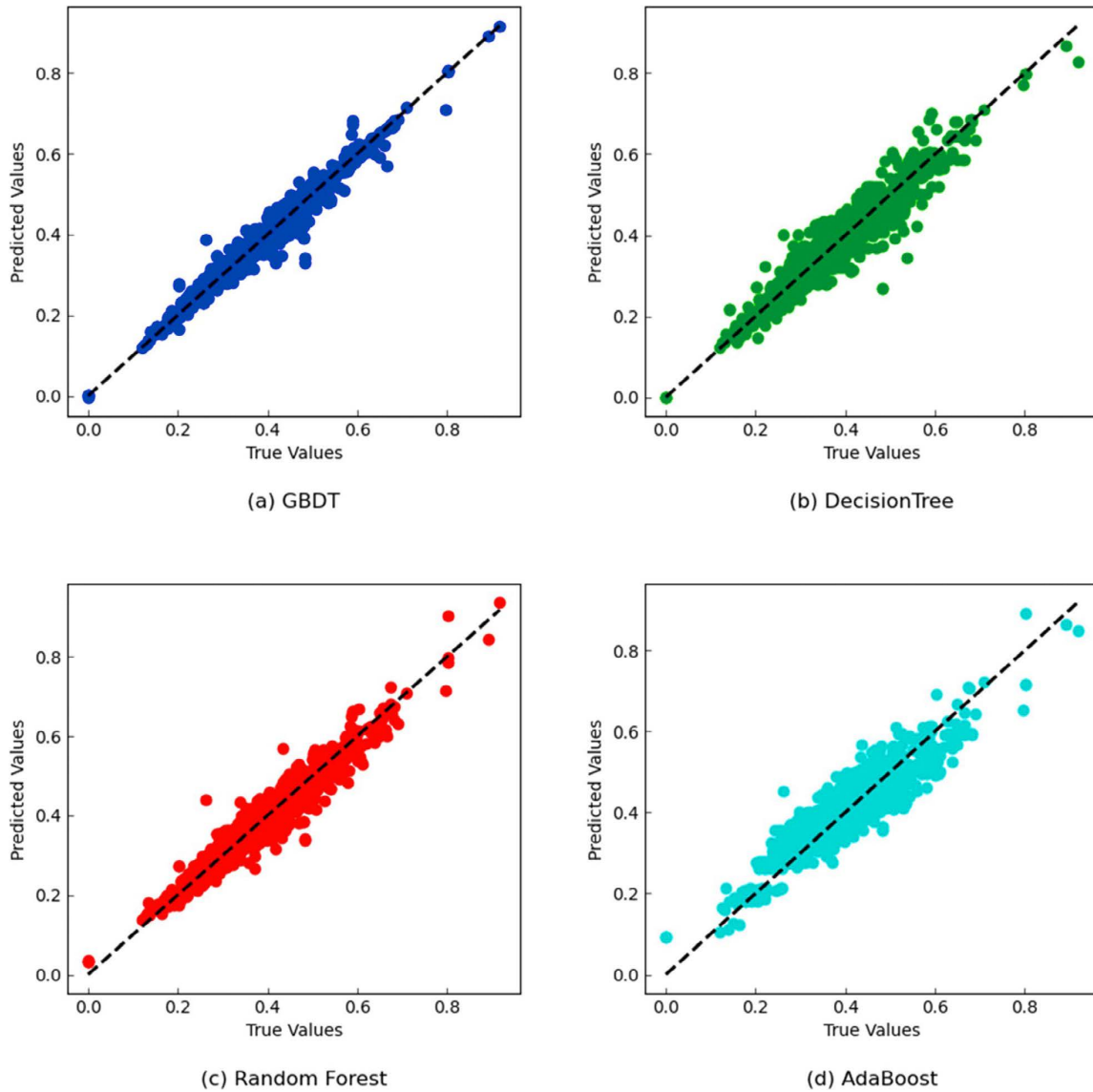


Fig. 10. Scatter plot of actual and predicted silicon content of hot metal. (Online version in color.)

The Adaboost (Adaptive Boosting) algorithm is an integrated learning algorithm designed to build a strong regressor by combining multiple weak regressors.³³⁾ The Adaboost algorithm is able to progressively improve the regression performance of the model by iterative training and adjusting the weights of the samples, which results in a combination of weak regression models with stronger regression capabilities. According to the Optuna framework, the number of base weak learners is 240, and the contribution scaling factor of each base learner is about 0.184.

6.3.2. Analysis of Experimental Results

Fifty sets of data were randomly selected from the test set to predict the silicon content of hot metal using Optuna optimized GBDT, Decision Tree, RF, and Adaboost algorithms, and the results are shown in **Fig. 9**.

Figure 10 presents the relationship between the predicted and actual values of the silicon content of hot metal by the different optimized algorithms on the whole test set. The correlation between the predicted and actual values can be observed through the scatter fit plot. If the data points are closer to a perfect diagonal line (with a slope of 1), the pre-

dicted values are highly consistent with the actual values, whereas if the data points are farther away from the diagonal line, it indicates that there is a large discrepancy between the predicted and actual values.

A comparison of the performance metrics of the different algorithms is shown in **Table 5**. It can be seen that compared to the other 3 algorithms, the GBDT algorithm presents the best results, with the coefficient of determination R^2 closest to 1, and the RMSE and MAE are also minimized.

Table 5. Different algorithm performance metrics.

Model	R^2	RMSE	MAE
GBDT	0.975	0.0152	0.0098
DecisionTree	0.919	0.0275	0.0200
RF	0.945	0.0227	0.0163
Adaboost	0.849	0.0295	0.0377

7. Conclusions

After preprocessing and feature screening of a large

amount of ironmaking data, the Optuna-GBDT model for prediction of silicon content of hot metal was constructed, and the following conclusions were drawn:

(1) Compared with the traditional random search method and Bayesian algorithm, the Optuna framework performs better in optimizing the GBDT model parameters. It can achieve the lowest error with the least number of iterations, showing its efficiency and superiority.

(2) The Optuna framework-optimized GBDT algorithm outperforms optimized algorithmic models such as Decision Tree, Random Forest, and AdaBoost in predicting the silicon content of hot metal. The mean absolute error (MAE) of the model is reduced to only 0.0094, the root mean square error (RMSE) is reduced to 0.0152, and the coefficient of determination (R^2) reaches 0.975, which indicates that the model is able to predict the silicon content of hot metal with a high degree of accuracy and provides reliable guidance for the ironmaking process.

(3) The description of the above results high-lights the advanced nature of the Optuna-GBDT model in terms of optimization process and prediction accuracy. This model not only optimizes the parameters efficiently but also provides a more accurate pre-diction of the silicon content of hot metal. This is of great significance to the ironmaking industry and can help improve productivity, reduce costs, and optimize product quality.

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REFERENCES

- Q. Li: Ph.D. thesis, Zhejiang University, (2005), <https://d.wanfangdata.com.cn/thesis/Y929223>, (accessed 2023-09-01).
- L. Jian, S. Gong and Y. Wang: *J. Zhejiang Univ. Sci. B*, **34** (2007), 282 (in Chinese). <https://doi.org/10.3321/j.issn:1008-9497.2007.03.010>
- L. Shi, Z. Li and G. Cui: *J. Inner Mongolia Univ. (Nat. Sci. Ed.)*, **41** (2010), 4 (in Chinese). <https://doi.org/10.3969/j.issn.1000-1638.2010.04.013>
- T. Yu, J. Li, M. Li and L. Shi: *J. Inner Mongolia Univ. (Nat. Sci. Ed.)*, **46** (2015), 5 (in Chinese). <https://doi.org/10.13484/j.nmgdxzbk.20150516>
- J. Zhao, H. Li, X. Li, X. Liu, H. Li and Q. Lyu: *China Metall.*, **31** (2021), 93 (in Chinese). <https://doi.org/10.13228/j.boyuan.issn1006-9356.20200701>
- X. Liu, Y. Deng, X. Li, L. Hao, E. Liu and Q. Lyu: *China Metall.*, **31** (2021), 10 (in Chinese). <https://doi.org/10.13228/j.boyuan.issn1006-9356.20200391>
- Z. Li, C. Yang, W. Liu, H. Zhou and X. Li: *CIESC J.*, **69** (2018), 6 (in Chinese). <https://doi.org/10.11949/j.issn.0438-1157.20171534>
- L. Jian, C. Gao, L. Li and J. Zeng: *ISIJ Int.*, **48** (2008), 1659. <https://doi.org/10.2355/isijinternational.48.1659>
- J. Yang, H. Wang, X. Wang and L. Zhang: 3D Imaging—Multidimensional Signal Processing and Deep Learning, Springer Nature Singapore, Singapore, (2022), 45. https://doi.org/10.1007/978-981-19-2452-1_5
- Z. Cui, A. Yang, L. Wang and Y. Han: *Metals.*, **12** (2022), 1403. <https://doi.org/10.3390/met12091403>
- A. P. M. Diniz, K. F. Côco, F. S. V. Gomes and J. L. F. Salles: *Metals.*, **11** (2021), 1001. <https://doi.org/10.3390/met11071001>
- H. Zhang, S. Zhang, Y. Yin and X. Chen: *Int. J. Mach. Learn. Cybern.*, **9** (2018), 1697. <https://doi.org/10.1007/s13042-017-0674-8>
- H. Li, X. Li, X. Liu, X. Bu, H. Li and Q. Lyu: *Ironmak. Steelmak.*, **49** (2022), 283. <https://doi.org/10.1080/03019233.2021.1992816>
- Y. Fang, X. Zhao, P. Zhang, L. Liu and S. Wang: *Control Theor. Appl.*, **37** (2020), 1644 (in Chinese). <https://doi.org/10.7641/cta.2020.90571>
- S. Liu, Q. Lyu, X. Liu, Y. Sun and X. Zhang: *ISIJ Int.*, **59** (2019), 2156. <https://doi.org/10.2355/isijinternational.ISIJINT-2019-059>
- T. Akiba, S. Sano, T. Yanase, T. Ohta and M. Koyama: Proceedings of the Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, ACM, Anchorage, AK, (2019), 2623. <https://doi.org/10.1145/3292500.3330701>
- H. Li, X. Liu, X. Li, H. Li, X. Bu, S. Chen and Q. Lyu: *ISIJ Int.*, **62** (2022), 2301. <https://doi.org/10.2355/isijinternational.ISIJINT-2022-037>
- H. Li, X. Bu, X. Liu, X. Li, H. Li, F. Liu and Q. Lyu: *ISIJ Int.*, **61** (2021), 108. <https://doi.org/10.2355/isijinternational.ISIJINT-2020-249>
- I. Pratama, A. E. Permanasari, I. Ardiyanto and R. Indrayani: ICITSI, IEEE, Bandung, Indonesia, (2016), 1. <https://doi.org/10.1109/ICITSI.2016.7858189>
- J. Zhao, S. F. Chen, X. J. Liu, X. Li, H. Y. Li and Q. Lyu: *Int. J. Miner. Metall. Mater.*, **28** (2021), 1001. <https://doi.org/10.1007/s12613-021-2301-7>
- D. F. Williamson, R. A. Parker and J. S. Kendrick: *Ann. Intern. Med.*, **110** (1989), 916. <https://doi.org/10.7326/0003-4819-110-11-916>
- P. Trebuňa, J. Halčinová, M. Fil'o and J. Markovič: Proceedings of the 12th SAMI, IEEE, Herl'any, Slovakia, (2014), 381. <https://doi.org/10.1109/SAMI.2014.6822444>
- Y. Deng and Q. Lyu: *ISIJ Int.*, **60** (2020), 898. <https://doi.org/10.2355/isijinternational.ISIJINT-2019-545>
- J. Benesty, J. Chen, Y. Huang and I. Cohen: Noise Reduction in Speech Processing, I. Cohen, Y. Huang, J. Chen and J. Benesty, Springer, Berlin Heidelberg, (2009), 1. https://doi.org/10.1007/978-3-642-00296-0_5
- K. Jacobs: Discrete Stochastics, K. Jacobs, Birkhäuser, Basel, (1992), 155. https://doi.org/10.1007/978-3-0348-8645-1_7
- A. Giffin: *Phys.*, (2009). <https://doi.org/10.48550/arXiv.0901.2987>
- Z. Cai, Y. Long and L. Shao: *Pattern Recogn. Lett.*, **125** (2019), 396. <https://doi.org/10.1016/j.patrec.2019.05.021>
- B. Efron: *Science*, **340** (2013), 1177. <https://doi.org/10.1126/science.1236536>
- Y. Y. Song and Y. Lu: *Shanghai Arch. Psychiatry.*, **27** (2015), 130. <https://doi.org/10.11919/j.issn.1002-0829.215044>
- H. Shoji, Y. Hamada and A. Inoue: *ISIJ Int.*, **19** (2020), 227. <https://doi.org/10.5057/ijae.TJSKE-D-20-00030>
- Y. Mishina, R. Murata, Y. Yamauchi, T. Yamashita and H. Fujiyoshi: *IEICE Trans. Inf. Syst.*, **E98.D** (2015), 1630. <https://doi.org/10.1587/transinf.2014OPP0004>
- S. Osawa, T. Miyoshi and P.-j. Chun: *ISIJ Int.*, **2** (2021), 1. https://doi.org/10.11532/jsceiii.2.1_1
- K. W. Walker and Z. Jiang: *J. Acad. Librariansh.*, **45** (2019), 203. <https://doi.org/10.1016/j.acalib.2019.02.013>